CPU Thermal Cooling Fins

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Abstract

This study investigates the thermal performance of CPU heat sinks by evaluating the impact of both material selection and fin geometry on heat dissipation efficiency. Using ANSYS Multiphysics simulation, four materials (Stainless Steel (304), Brass Leaded, Aluminum, and Copper) were tested in combination with 3 distinct pin-fin geometries (rectangular, circular, and triangular. A 2D CFD simulation analysis was needed to validate the convective heat transfer coefficient (16.844 W/m^2*K) against academic standards, followed by 3D steady-state thermal and fluid flow simulations. The results demonstrate that thermal conductivity (k) is the primary influence for performance, with Copper consistently providing the lowest base temperature. Furthermore, geometry analysis showed that rectangular fins outperformed circular and triangular designs within each material by maximizing the surface area for the air-solid interaction. The optimal configuration was the Copper with rectangular fins, that achieved a minimum base temperature of 306.7 K. This is a 51.9 K reduction compared to the least effective Stainless Steel-Triangular setup. These findings support and confirm that maximizing convective surface area in high conductivity materials is crucial for optimizing CPU cooling in high-performance computer processing environments

Introduction

CPUs in computers generate large amounts of heat when performing computational tasks. Heat sinks are used to help dissipate the thermal energy and keep the CPU cool to avoid damage and slowdowns. To optimize the design of heat sinks, the fin geometry and material were experimented with. These metrics allow designers of heat sinks to be able to decrease the

temperature of the CPU. Through testing circular, rectangular, and triangular inline pin-fin geometries, the rectangular geometry will yield a higher convective heat transfer coefficient (h) in the 2D computational fluid dynamics simulation. The convective heat transfer coefficient is calculated by the formula

$$Q=h\bullet A(T_{Surface} - T_{Fluid})$$

$T_{Surface}$ is the surface temperature in Kelvin. T_{Fluid} is the fluid temperature in Kelvin. A is surface area in meters, Q is convective heat transfer rate in $W/m^2 \bullet K$.

As well as the copper version of each geometry will outperform, the other materials being Aluminum, Brass Leaded, Copper and Stainless Steel (304).

The thermophysical properties of the selected materials help determine heat transfer performance.

Table 1 [9]: Materials and their respective thermal conductivity, material density, and specific heat capacity from Thermtest[9]

Material	Thermal Conductivity (k) [W/m•k]	Density (ρ) [kg/m³]	Specific Heat (c_p) [J/kg•K]
Aluminum	225.94	2698	921
Copper	397.48	8940	385
Brass Leaded	117.15	8500	377
Stainless Steel (304)	14.644	7920	502

As shown in Table 1, Copper has the highest thermal conductivity of 397.48 W/m.K, making it the most effective material for conducting heat. Whereas Stainless Steel (304) has the lowest thermal conductivity of 14.633 W/m.K indicating poor heat transfer capability. Aluminum offers a significant advantage in terms of weight due to its low weight density of 2698 kg/m^3, this makes it a practical choice for most real-life applications. Specific heat capacity values vary

between 377 J/kg•K for Brass to 921 J/kg•K. This determines the amount of energy needed to raise the temperature of the material, influencing its thermal response under operating conditions.

This was tested using ANSYS simulations with the necessary physics and boundary conditions. A heat sink sat atop a CPU that is pulling heat away from the CPU. The heat sink had varied fin shapes and fin materials to study which combination was able to dissipate thermal energy best. An initial 2D simulation was setup with heat flux being applied to the base of the model and convection being applied to all sides of the heat sink except for the bottom. The simulation was run with energy enabled to look at convective heat transfer coefficients to compare values to the values from the Engineering Toolbox[10], which is an academic source for coefficients and engineering related values. The values obtained were inline from what was published in the source. The obtained value for the convective heat transfer coefficient (h) was then used in the 3D simulations. Both 2D and 3D simulations involved boundary conditions of the base of the heat sink sitting on top of the CPU receiving the heat flux and all other sides except the base experienced convection to let the heat out of the model. The model themselves was also enclosed in a large, closed area with velocity coming from the inlet on the left and traveling through the heat sink the moving towards the outlet on the right side of the enclosure. The outlet measured the pressure of the velocity.

The physics present in the simulations were thermal dynamics and fluid mechanics. The thermal dynamics were used in the use of heat flux and convection in the model. This allowed the heat to flow through the model itself and go from the CPU through the heat sink and out the fins. This behavior being similar to how a heat sink in a computer thermally dissipates the heat. Midplane temperature was measured to see which geometry and material had the lowest temperatures. The fluid mechanics present were in the air flowing over the heat sink fins to

simulate how air moves inside a computer case. Coming from an inlet fan, blowing across the motherboard and CPU, then flowing towards the outlet of the case. This study differed from studies of the past due to the inclusion of studying both fin geometry and material at the same time. Previous studies only focused on one of the two parameters. Additionally, the other studies only used flat plate fin designs [2]. Our studies included differing types of fin designs such as rectangular, triangular, and rounded shapes.

The simulations themselves consisted of attempting to create a heat sink design that dissipates heat most effectively by varying the geometry and materials used to find the best combination. The CPU was modeled as a constant heat source which applied thermal energy (heat flux) to the bottom of the heat sink itself. This thermal energy then moved through the heat sink and dissipates out of the sides and fins of the heat sink using the principal of convection. Airflow as modeled to flow past the fins, simulating a computer case fan. This helped increase dissipation of heat, similar to how it is in real life. The various materials used had the exact same boundary conditions between simulations. The simulations produced temperature distributions, airflow velocity profiles, and pressure data which were compared to provide a clear insight on the most effective heat dissipation design.

Scientists and engineers would be interested in this problem of keeping CPU temperatures low with more efficient heat sinks because innovations in this field better help them with their areas of research. Computational power of a CPU is directly correlated with how well the heat can be dissipated. When a CPU's temperature reaches too high of a value, thermal throttling occurs which means that the CPU isn't able to process as much data as fast as it ideally could. By keeping the temperatures low this problem never arises and allows more powerful CPUs to be made. Because as computational ability increases, so does the thermal energy

generate by the CPU. Scientists and engineers use CPUs to be able to perform complex simulations and data analysis with programs such as ANSYS and MATLAB. The time in which these programs take to run heavily depends on CPU data processing speeds. If technology advances further in heat sink design, then new CPUs can push the envelope of data processing even further. These innovations then trickle down to the general public using computers as well. Everyone wants to be able to quickly look up something on the web or get an AI output quickly. By having more powerful CPUs in general consumer electronics, this allows society as a whole to have access to faster computing technology. The wait times for something to load on a computer or smartphone will get greatly reduced as CPUs are able to be more powerful.

While researching the project, many important concepts were discussed in academic sources and YouTube videos. The geometric design of a heat sink is a very important factor in heat dissipation since the choice of fin cross-section will directly impact the surface area. In recent studies, circular pin fins have been shown to be superior in heat transfer compared to its counterparts like conical geometry due to an increase in area. Beyond the shape along, the arrangement of the pins either being inline or staggered also affects heat transfer. Inline arrays are easier to manufacture, but they have a downside of an increase in airflow resistance. Conversely, staggered arrays increase the spread and path of the fluid, which enhances convection heat transfer. However, this has a downside of increased pressure drop across the heat sink and more complex manufacturing[1]. The choice of material is another important part for thermal management, as the thermal conductivity (k) will decide the rate of heat transfer from the CPU to the heatsink. While there are various metals and alloys used for CPU cooling, there are tradeoffs between thermal performance, manufacturing complexity, and cost. Aluminum is a very common choice due to its weight to strength ration and affordability; however, the thermal

conductivity is lower than more expensive options like copper. Analytical models suggest that utilizing materials with higher thermal conductivity have led to a reduction in surface temperature and thermal stress, which is very important for stability under intense temperature gradients[2]. 3D modeling is essential for modern thermal analysis to evaluate complex geometries before creation. The geometry of the heatsink is typically made into a mesh, to be able to solve governing fluid flow and energy equations. While using a coarse mesh will be able to analytically solve it faster, it will be less accurate. Consequently, a fine mesh is crucial for obtaining accurate results and is needed to evaluate high power components like a computer's CPU to ensure the utmost accuracy[3]. The analysis of threat sink performance typically relies on steady-state physics. This approach simplifies complex physics and computational complexity, which allows for a focus and emphasis on the equilibrium state of the design. The physics and equations used include continuity, Navier-Stokes, and energy equations. These equations represent mass conservation, balancing forces with moving fluids, and the transportation of thermal energy between the solid fins and the cool air[4]. The overall performance of the heat sink is a combination of air flow dynamics, and the pressure drops across the fin array. Convective heat transfer is from the movement of the air between the spacing of the fins as it travels through it. The inlet velocity as well as the characteristic fluid properties are major constants that will determine the convective heat transfer. As the air travels through the fin array, the fins act as obstructions that will create resistance that will show a reduction in pressure from the inlet to the outlet. An increase in inlet velocity will enhance the heat transfer coefficient and lower the surface temperature, as well as increase the pressure losses. An optimal design is where the cooling efficiency is maximized without exceeding the pressure limits of the fan or the system[5].

The three video tutoritals[6-8] serve as references and tutorials for the ANSYS simulations and provide numerous ways to model the system. These include techniques for visualizing velocity spread[6] and temperature spread[6-7], as well as structural analysis to evaluate thermal stress and deformation where necessary[8].

Methods/Experimental Design

The methods used in the ANSYS simulations were fluid mechanics and thermodynamics. Fluid mechanics were used to simulate airflow over the fins. This allowed the convective heat transfer to be modeled. The airflow was treated as a steady state and incompressible flow with computational fluid dynamics being used to solve the model in ANSYS. The velocity inlet was on the left side of the enclosure, and the pressure outlet was on the right side of the enclosure. Both of these boundary conditions were implemented to allow the airflow to have somewhere to start from and end. The airflow across the heat sink interacts with it and creates velocity gradients. This effect helped increase the heat transfer efficiency of the heat sink. The vector velocity contours, streamlines, and pressure gradients were analyzed to help understand how the different fin geometries and materials influenced the airflow and cooling of the system.

The other physics being replicated in the simulations was thermodynamics. Thermodynamics were used to analyze the amount of heat transfer from the CPU through the heat sink and out into the air around. More specifically, heat conduction through a solid material was used along with convective heat transfer between the heat sink and the surrounding airflow used. The CPU was modeled as a constant heat source and produced a heat flux that was applied to the bottom of the heat sink itself. The rate of conduction depended on which specific material was used for a given simulation. At the interface between the solid surface of the heat sink and the air, convective heat transfer occurred and was influenced by the airflow characteristics. Additionally, temperature

distributions across the CPU were modeled with the heat sink baser and fins being analyzed using contour plots in FLUENT (Built in Solution tool in ANSYS). The key thermal metrics that were looked at were maximum and minimum temperatures, temperature gradients, and midplane temperatures.

Some simplifications that were made were using steady-state analysis in ANSYS. This got rid of the complex effects such as startup heating and time dependency. It also helped the focus remain on long term thermal equilibrium. Another aspect that was not used was radiation heat transfer because the dominant heat transfer methods in the model were only conduction and convection. Simplified geometry allowed less computer resources to be used so that computational solving time was kept to a reasonable level for this scope of project. The airflow was also treated as incompressible to help reduce computational difficulty while still having the core structure and physical behavior of the system.

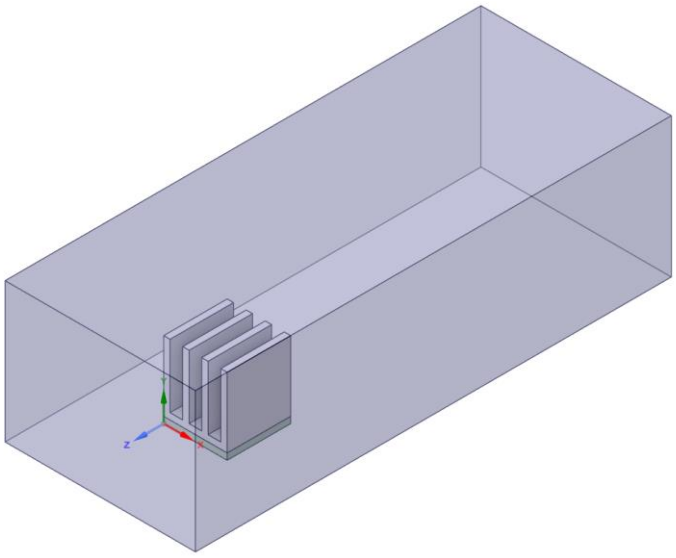


Figure 1: 3D geometry of CPU with Heat Sinks surrounded by an Enclosure made in Ansys SpaceClaim.

Both 2D and 3D geometry were used in the ANSYS simulations to capture the behavior of heat transfer and airflow of the system. A 2D computational fluid dynamics simulation was used to analyze the cross-sectional fluid (air) over the rectangular, circular, and triangular fin shapes. This was used to calculate the convective heat transfer coefficient (h). A 3D simulation was then used by extruding the existing 2D geometry consisting of the rectangular CPU base. The heat sink base was then mounted directly (in-contact) with the CPU to allow for conductive heat transfer to occur. The cooling fins of the heat sink were then extruded sketches in an inline array to maximize surface contact area with the airflow. The mesh used for the 3D simulation is shown in Figure 2, where the mesh was refined in regions near the fin surfaces to capture thermal gradients and airflow results, which was able to provide better simulation results.

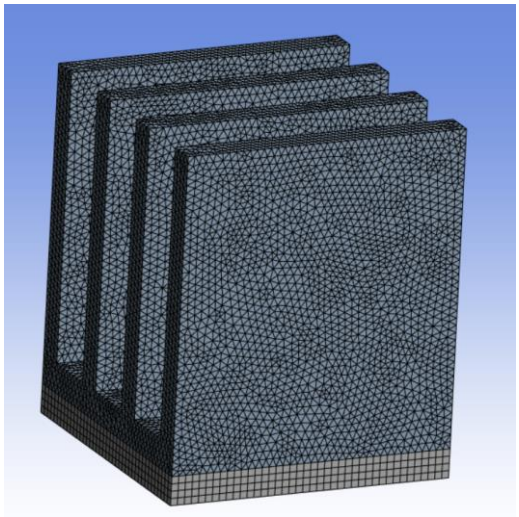


Figure 2: Mesh at the Heat Sink Fins

The boundary conditions in Figure 1 used consisted of a velocity inlet (on the left side of the enclosure) that was defined with a constant air velocity of 1m/s to simulate cooling from a

computer fan. A pressure outlet (on the right side of the enclosure) was set to zero-gauge pressure to allow the air to exit the system while we were able to measure the pressure drop. Another boundary condition used was a constant heat flux applied to the base of the heat sink to simulate the heat being generated from a CPU working. All surfaces of the heat sink itself other than the bottom experienced convection to allow the heat to dissipate out of the system. Additionally, a no slip condition applied to all fluid-solid interfaces to model the velocity boundary layers.

Results and Discussion

Convergence Check

Mesh convergence was achieved by analyzing the effect of mesh refinement on the maximum temperature of the heat sink. This was done by plotting the number of elements of the mesh vs the max temperature of the heat sink. As the mesh density increased, the variation in maximum temperature decreased and eventually converged. Conversion was achieved at **65,579 elements,** where the maximum temperature was at 309.87 K. The maximum temperature corresponds to approximately 36.72°C. This falls within the typical operating temperature range of CPU cooling systems.

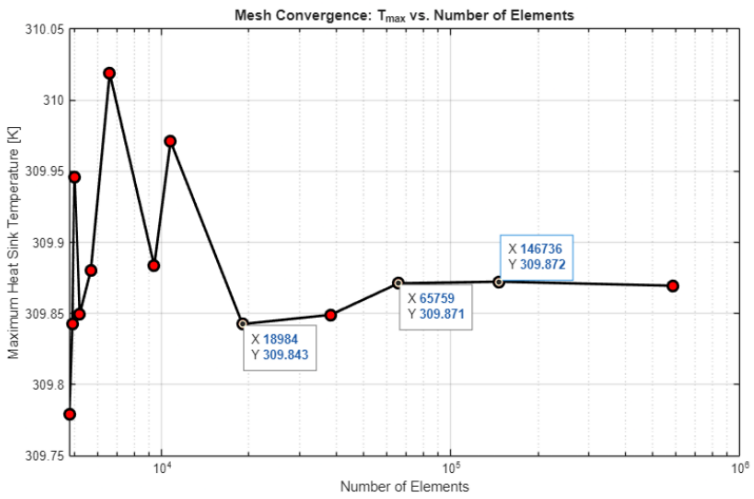


Figure 3: Mesh Convergence Study. Plot of maximum temperature vs. number of elements. The solution converges at around 65,759 elements.

A timestep of convergence was not required as the simulations were conducted under steady-state conditions. In a steady-state model, most systems and equations run without any time dependence. Based on attempts to perform a geometry convergence study, variables such as fin height and heat sink base thickness were considered. The goal was to evaluate these effects by plotting the Heat Transfer Coefficient (h,W/m^2•K) and Heat Transfer Rate (Q, W). However, convergence was not successful due to technical limitations within Ansys Workbench. Specifically, when the features were parametrized in the Design Modeler, the named selections were replaced causing Fluent to lose references. This resulted in simulation errors. Despite these limitations, a value for h was obtained when it was solved for convergence using the initial conditions.

$$h = 16.844 \text{ W/m}^2 \text{•K}$$

This value falls within the range specified for forced convection in air (10-1000 W/m^2*K). This range was obtained from engineeringtoolbox [10]. The value Fluent calculated

falls within these boundaries, which was deemed accurate for use in subsequent 3D simulations and analysis.

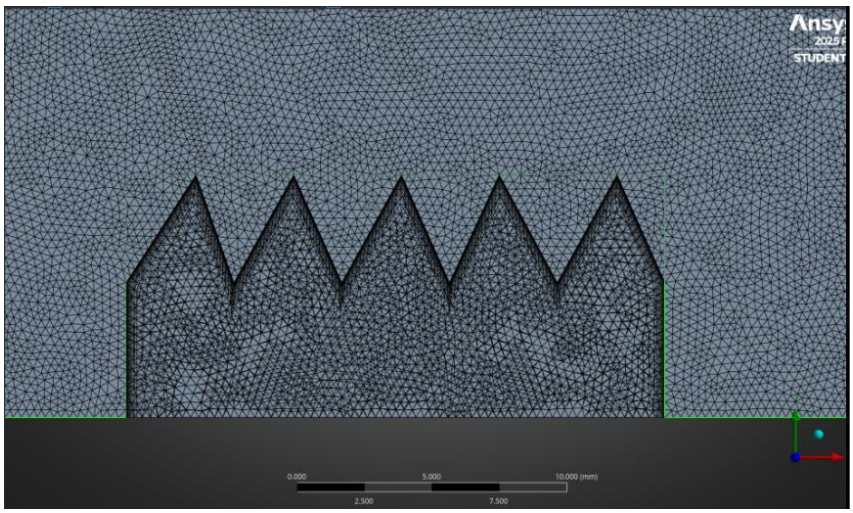


Figure 4: Mesh for triangular fin geometry in 2D CFD simulation

The 3D temperature contours of the entire system illustrate the steady-state thermal distribution through the entire model and surroundings. The highest temperature is observed at the CPU base where a constant volumetric heat source value of 10,000,000 W/m^3 was used. This value was calculated based on 1W of power consumed along with the geometry of the entire model. Heat is conducted from the base into the fins and dissipated into the surrounding air through convection.

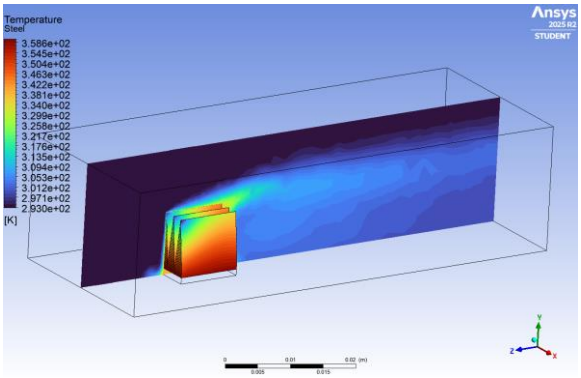


Figure 5: 3D Steady State Temperature Contour of the Stainless Steel Heatsink with Triangular Pin-Fin Array

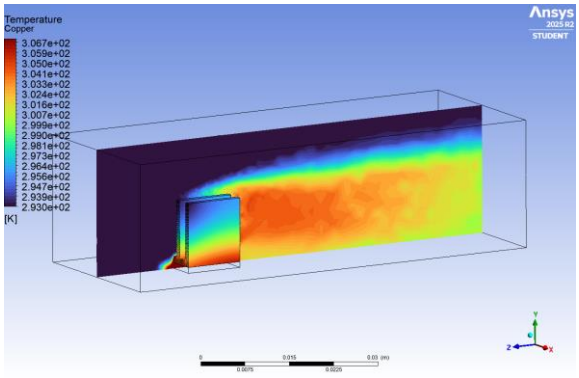


Figure 6: 3D Steady State Temperature Contour of the Copper Heatsink with Rectangular Pin-Fin Array

A clear temperature gradient is observed along the length of the fins, indicating effective heat transfer from the base to the fin tips. Materials with higher thermal conductivity, such as copper and aluminum, show a more uniform and rapid decrease in temperature along the fins. Whereas stainless steel and brass retain higher temperatures near the base, indicating greater thermal resistance.

The velocity contours provided a better reference to the airflow behavior over the heat sink. Triangular fins allow for a smoother airflow but exhibit slightly reduced convective performance. Whereas rectangular and circular fin geometries tend to enhance airflow interaction and provide turbulence near the surface, improving heat transfer.

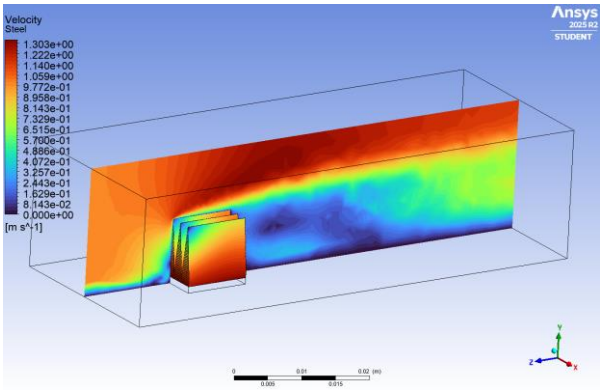


Figure 7: 3D Steady State Velocity Contour of the Stainless Steel Heatsink with Triangular Pin-Fin Array

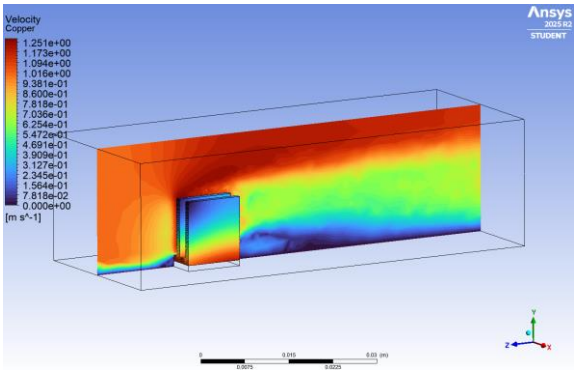


Figure 8: 3D Steady State Velocity Contour of the Copper Heatsink with Rectangular Pin-Fin Array

A clear trend is observed with respect to material selection. As the thermal conductivity of the material increases, base temperature decreases. Stainless steel exhibits the highest temperature followed by brass, aluminum, and copper, which consistently provides the lowest temperature for all fin geometries. Fin geometry also plays a significant role in heat dissipation. Across all materials, rectangular fins resulted in the lowest base temperature, indicating the best cooling performance. Circular fins provide moderate performance, and the triangular fin geometry consistently showed high temperatures.

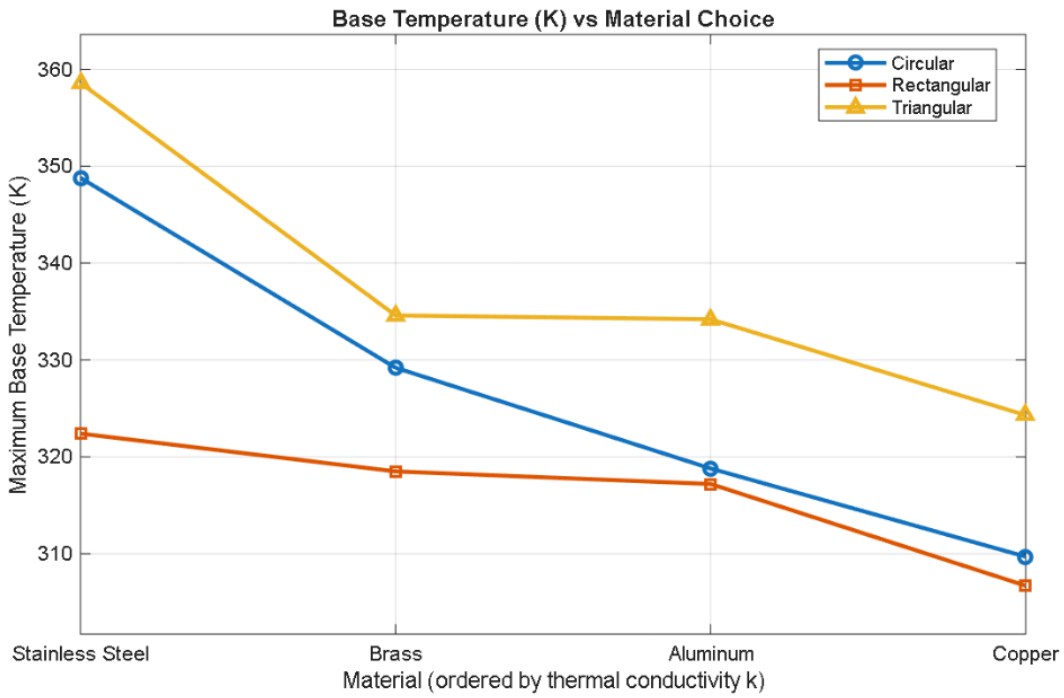


Figure 9: Maximum Base Temperature (K) vs Material Choice for Various Fin Geometries. The plot displays the relationship between the thermal conductivity (k) and its cooling performance for circular, rectangular, and triangular fin arrays. Materials are ordered on the x-axis from the lowest to highest thermal conductivity (Stainless Steel to Copper).

As shown in Figure 9, the simulation results provide a direct comparison of how the base temperature responds to changes in both the material and geometry. The data shows a clear inverse relationship, so as thermal conductivity (k) of the material increases, the maximum base temperature decreases. For example, moving from a low-conductivity material like Stainless Steel to a high-conductivity material like Copper results in a stark reduction in temperature regardless of the fin shape used. Specifically, the Stainless Steel-Triangular configuration recorded the highest temperature of 358.6 K. The configuration that had the lowest recorded temperature was the Copper-Rectangular with a value of 306.7 K. This reflects a total temperature difference of 51.9 K. Within each material category, the rectangular fins consistently

maintained the lowest temperature curve, followed by the Circular fins, and then lastly the Triangular fins. These results confirm that while material properties are a dominant factor in heat transfer, the fin geometry plays a critical role too. Materials wise, Copper's high k value (397.48 W/m*k) allows it to conduct the thermal energy away from the CPU faster than the Stainless Steel (14.644 W/m•k). Geometrically, the superior performance of the rectangular fins suggests that maximizing the surface area for air-solid interface is more effective for cooling than more aerodynamic benefits that come from triangular or circular shapes. These findings support the initial hypothesis and suggest that for optimal CPU cooling, a high-conductivity material must be paired with a geometry that maximizes its convective surface area.

Conclusion

The study investigates the thermal performance of a CPU with a heat sink by analyzing the effects of fin geometries along with material selections using Ansys simulations. An organized and systematic approach was taken to evaluate the three fin geometries: rectangular, circular, and triangular, and the four materials: stainless steel, brass, aluminum, and copper.

Mesh convergence was successfully achieved, making sure the entire model accurately depicts a real-world system. A steady-state approach was used, eliminating the need for a timestep.

The results demonstrate that materials with higher thermal conductivity consistently resulted in lower base temperature, with copper providing the best performance and stainless steel the worst. Similarly, fin geometry played an important role with rectangular fins outperforming both the circular and triangular designs due to increased surface area.

The optimal configuration identified was a copper heat sink with rectangular fins which achieved the lowest base temperature. It provided effectiveness of up to 50 K.

Overall, the findings confirm the initial hypothesis that both fin geometry and material selection directly impact cooling performance. This study highlights the core fluid concepts of conduction and convection mechanisms in thermal system design. These findings are applicable to real-world CPU cooling systems and impact how they can further be optimized.

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